

RIDA MELKAOUI

INDUSTRIAL ENGINEER | DATA, AI & BUSINESS INTELLIGENCE

ridamelkaouiofficial@gmail.com | +212 620 999 885 | Kenitra, Morocco

LinkedIn | GitHub | Portfolio



PROFILE

Industrial & Production Engineer who combines supplier-quality ownership with data analytics, BI and applied AI. At Magna, leads electronics and metal supplier quality across international programs and builds the operating views used to track launch readiness, issues and actions. Translates industrial problems into measurable decision products with Python, SQL and Power BI.

PROFESSIONAL EXPERIENCE

MAGNA INTERNATIONAL <i>Supplier Quality Engineer — Electronics & Metal Commodities</i>	Kenitra, Morocco Mar 2026 — Present
- Own supplier-quality performance for electronics and metal commodities across China, South Korea, Spain, Slovakia, Germany and Morocco, using launch readiness, 8D/CAPA status and open-issue risk to focus escalation. - Developed X-Ray, a full SQE BI and workflow system covering nomination, Safe Launch and SOP; unified ownership, milestones, PPAP/DVP evidence, critical parts and open issues across 6 control modules and 4 lifecycle gates. - Implemented KPI views and an automated notification pipeline for missing baselines, overdue actions and gate risks, keeping responsibilities and deadlines visible across the supplier portfolio.	
STELLANTIS R&D <i>Engineering Intern — Data Analytics, Process Optimization & AI</i>	Casablanca, Morocco Feb 2025 — Aug 2025
Mapped the E-BOM/M-BOM anomaly process and built a Python application plus structured Databricks model for detection, prioritization and traceability; reduced the validated treatment cycle from 2 days to under 3 minutes (−98% manual handling, 100% project test-case reliability).	
SANY — FORGE DE BAZAS <i>Engineering Intern — Process Digitalization & Maintenance</i>	Casablanca, Morocco Jul 2024 — Sep 2024
Built a Unity/NLP assistant that structured access to maintenance and spare-parts knowledge, replacing fragmented shop-floor search with a guided information flow.	
3D SMART FACTORY <i>Data Analyst Intern</i>	Mohammedia, Morocco Aug 2023 — Sep 2023
Structured Power BI and Excel reporting around OEE, FPY, scrap and downtime, giving Lean reviews a single view of production loss drivers and bottlenecks.	

PROJECTS

Demand / Order — Forecasting & replenishment
Created to answer what to replenish and where to review exceptions, this cutoff-safe Python engine models 900 M5 item-store series, challenges forecasts against calibration-selected seasonal baselines and translates the winning signal into inventory-policy scenarios. It improved WAPE by 4.5% and simulated 1,528 fewer lost units at 96.7% fulfilled demand on a 28-day holdout.
Route / Control — Optimization & operations
To help dispatchers balance time windows with coherent routes, learned zone-order preferences from 6,112 public histories, combined directed travel times with constrained local search, and surfaced exceptions in a control tower. On a separate 13-route cohort, the policy cut simulated drive time 2.8%, kept 100% timed-package adherence and removed 332 zone re-entries.
Agent / Proof — AI evaluation & observability
Built an evaluation lab to decide whether a tool-using AI agent is repeatable enough to release, analyzing 3,336 official trajectories across pass@k, repeatability, cost, latency and failure modes. The pinned GPT-4.1 telecom cohort exposed a 14.9-point gap between one-shot and repeatable success.

EDUCATION

Mohammed VI International Civil Aviation Academy (AIAC)	Nouaceur, Morocco
Engineering Degree — Industrial & Production Engineering	2022 — 2025
Salmane El Farissi Preparatory Classes	Salé, Morocco
MPSI — MP	2020 — 2022

SKILLS

DATA & ANALYTICS	Python · SQL · Pandas · NumPy · scikit-learn · statistics · forecasting · experimentation · data modeling · Databricks
BI & DELIVERY	Power BI · DAX · Excel · Power Query · Tableau · React · TypeScript · FastAPI · Git · Docker · KPI systems · automation
INDUSTRIAL & QUALITY	Supplier/process performance · PPM · OEE · FPY · 8D/CAPA · FMEA · root-cause analysis · Lean · launch/SOP governance

LANGUAGES

Arabic — Native French — Fluent English — C1 (TOEIC 855; EF SET 62/100)

CERTIFICATIONS

IBM Data Analyst Professional Certificate IBM / Coursera · 2026	EF SET English Certificate — C1 EF SET · 62/100	TOEIC Listening & Reading — C1 Score 855
--	--	---